

Posters: Multicellular, Consortium, and Tissue Modeling

1539-Pos

Revealing key mechanisms of atrial fibrillation-associated fibrosis through computational modeling of cardiac fibroblast

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Atrial fibrillation (AF), the most common cardiac arrhythmia, is characterized by irregular atrial electrical activity, resulting in asynchronous atrial contraction, and associated with significant morbidity and mortality. The rapid atrial rates of AF contribute to substantial ionic and structural atrial remodeling, including fibrosis, which affects AF maintenance and the success of treatment strategies. Atrial fibrosis, whether in patches or diffuse, can disturb electrical conduction and cell-to-cell coupling and promote reentry or ectopic activity. Atrial fibroblasts, the principal fibrosis-driving cell types, are regulated by a complex signaling network, but the precise mechanisms through which this network orchestrates fibrosis formation in AF are yet to be elucidated. Indeed, inhibition of individual pathways has proven insufficient to completely normalize atrial fibrosis and AF duration, likely because alternative pathways can compensate for the absence of one. To quantitatively interrogate the fibrosis mechanisms involved in AF, we leveraged computational frameworks of ventricular fibroblast signaling and expanded these models to develop the first atrial signaling network model. This includes new atrial-associated genes/nodes involved in fibrogenesis and differentially expressed in chronic AF (vs normal sinus rhythm, nSR) patients and connections based on experimental evidence in fibroblast-related cell types. Model predictions for activation of fibrotic markers were validated against experiments conducted in human atrial Fbs (haFbs) exposed to different signaling contexts, with/without knockdown of network nodes. Model analysis revealed the most influential subnetworks/motifs and nodes that drive haFb activation/proliferation. Future work will integrate mRNA profiles of haFbs from AF vs nSR patients for optimizing our network model parameters to identify the nodes and signaling pathways that critically mediate haFbs activation/proliferation in AF vs nSR. Ultimately, this knowledge may guide the development of improved anti-AF therapies by preventing/reversing fibrogenic processes and atrial tissue remodeling.

1540-Pos

Meta-analysis of static magnetic field effects on blood and bone tissue: A comprehensive review

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The goal of this research is to provide a meta-analysis of multiple studies that delve into the impacts of static magnetic fields (SMFs) on blood and bone tissues. SMF exposures hold the promise of emerging as a minimally invasive therapeutic option for addressing diverse health concerns. This comprehensive review includes studies that were conducted within the last 20 years, using SMF strengths ranging from 5 nT to 16 T and exposure durations ranging from 15 minutes to 12 weeks. A variety of subject types were examined including mice, rats, hamsters, rabbits, and humans. SMF effects were found to depend heavily on SMF strength and duration. SMF exposures of 500 nT (weak) and 0.2 T (moderate) promoted osteoclast differentiation, formation and resorption, while 16 T (high) had an inhibitory effect on RAW 264.7 cells. Continuous chronic SMF exposure of mice at 60 mT for 7 days caused reduced venular diameters by 54.4% ($41.6 \mu\text{m} \pm 4.9\%$ vs. $91.2 \mu\text{m} \pm 8.3\%$, $p < 0.002$), reduced arteriolar diameters by 50% ($20.5 \mu\text{m} \pm 5.1\%$ vs. $40 \mu\text{m} \pm 4.2\%$, $p < 0.015$), and reduced venular functional length density by 60% ($33.3 \mu\text{m} \pm 5.7\%$ vs. $83.4 \mu\text{m} \pm 20.5\%$, $P < 0.025$). SMF exposure of 586 mT for 3 hours/day for 3 days resulted in retardation of tumor growth by 30%. Exposure of 30 - 80 mT SMF for 12 weeks significantly increased locomotor activity in rats ($P < 0.05$; $P < 0.001$), thereby increasing bone material density by 10%. This meta-analysis shows that exposure to SMF positively affects hemodynamics, blood pressure, tumor growth retardation, and bone healing and formation.

1541-Pos

Multicell modeling of the skeletal muscle microenvironment to explore age-related changes in satellite cell dynamics

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The regenerative potential of skeletal muscle tissue depends on cell-cell and cell-matrix interactions in the muscle microenvironment. Ageing muscle shows a

functional decline in stem cell dynamics [1]. Understanding the regulators of SC division in muscle fibre bundles is an important step in rescuing muscle regenerative potential. In silico modeling offers a high throughput method to investigate chemical and mechanical regulators of cell-cell and cell-matrix interactions. This preliminary study aims to recapitulate the effect of growth factors and cytokines, such as hepatocyte growth factor (HGF), on SC repair kinetics using multi-cell models of repair. Specifically, we evaluate the effect of increasing HGF concentration in recruiting SCs to the damage site. A 3D multi-cell model of the muscle microenvironment was created using CompuCell3D (CC3D), a virtual tissue modelling platform [2]. Agents were implemented in a $100 \times 10 \times 8$ lattice. Average SC diameter relative to muscle fibre diameter is represented in the model. Secretion of HGF from damaged tissue was varied from 0-50, and the time for SCs to reach the damage tissue location was recorded. Each time step represented one minute, and simulation end time was set to 3000 minutes. Initial SC distance from damage was set to ~ 20 pixels for all simulation conditions. Satellite cells did not reach the location of damage when HGF secretion was 10 or below. SCs time to reach damage location was decreased when HGF secretion from damaged tissue was increased. With secretion set to 50 or 40, distance to damage was comparable (~ 800 min). Secretion level of 30 took 1000 min to reach the damage location, while 20 took 1750 min.

1. Feige P et al. Cell Stem Cell. 2018 2. Swat MH et al. Methods in Cell Biology. 2012.

1542-Pos

Dynamic collagen matrix model for tissue simulations on distributed memory systems

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Recent research suggests that the extracellular matrix (ECM) has a significant influence on the development of cancer. However, most computational models of cancer pay little attention to the mechanical properties of the ECM. We introduce an elastic lattice model (ELM) as an extension to the cellular Potts model (CPM) and employ the Zener model of viscoelasticity to simulate the ECM. To seamlessly integrate the ELM with the CPM, we propose a flexible formalization of their interactions. The choice of the ELM is motivated by its reliance on local particle interactions, allowing us to implement it efficiently as a stencil code within cells in silico (CiS), our massively parallel CPM implementation. We conduct extensive computational experiments on both CPUs and GPUs, evaluating factors such as execution time, memory usage, and scalability. Finally, we assess the model's utility by attempting to replicate experimental findings, thereby demonstrating its potential in advancing our understanding of cancer development and the ECM's crucial role.

1543-Pos

In silico design of a microbial consortium for carbon-neutral butanol production

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Biological butanol production via microbial fermentation has become increasingly attractive due to concerns about rapid depletion of fossil fuels. However, traditional acetone-butanol-ethanol fermentation generates carbon dioxide (CO₂) as a byproduct and may contribute to global warming. Theoretically, a novel microbial consortium has the potential to synthesize biobutanol without net carbon emission, which involves three bacteria species: a lactic acid bacterium (*Lactococcus lactis*) that converts glucose to lactate, a mixotrophic acetogen (*Clostridium formicoaceticum*) that utilizes lactate and formate to produce acetate, and an engineered butanol-producing clostridia (*Clostridium tyrobutyricum*) that converts glucose and acetate to butanol. The byproduct CO₂ is converted to formate via electrochemical reactions and then reassimilated to the system. In this study, we use mathematical modeling to quantitatively analyze the consortium and to systematically search for design requirements that specify butanol production and CO₂ emission. Our study can provide a systematic *in silico* elucidation of the dynamic behaviors and metabolic characteristics of the butanol-producing, carbon-neutral consortium, thereby providing quantitative insights into microbial complex ecosystem metabolism and computational guidance to facilitate system development.

1544-Pos

How muscle solved the Gibbs paradox

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A thermodynamic analysis of mechanical systems i) further disproves the 17th century molecular power stroke mechanism, ii) accounts for many chemical, mechanical, and energetic aspects of muscle contraction, and iii) provides